MILITARY SPECIFICATION

PACKAGING OF DRIVE, MOTOR, M2A2

1. SCOPE

1.1 This specification covers the cleaning, preserving, packaging, packing and marking of Drive, Motor, M2A2 for each level of protection.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standard and drawing, of the issue in effect on date of invitation for bids, form a part of this Specification:

SPECIFICATIONS

Federal

FF-B-571	- Bolts; Nuts; Studs; and Tap Rivets.
FF-N-105	- Nails; Wire; and Staples.
FF-S-111	- Screws, Wood, Slotted-Head.
QQ-S-781	- Strapping, Flat; Steel.
QQ-S-790	- Strapping, Round; Steel, Bare and Zinc Coated.
UU-C-843	- Cushioning Materials, Cellulosic.
UU-T-111	- Tape; Gummed, Water-Resistant.
LLL-F-291	- Fiberboard; Corrugated, Single Face, (Flexible).
LLL-B-631	- Boxes; Fiber, Corrugated (for domestic shipment)

Military

JAN-P-100	- Packaging and Packing for Overseas Shipment - General Specification.
JAN-P-106	- Boxes; Wood, Nailed (Overseas Type)
JAN-P-108	- Boxes, Fiberboard; Corrugated and Solid (Military Type).
MIL-P-116	- Preservation, Methods of.
JAN-P-117	- Packaging and Packing for Overseas Shipment - Bags, Interior Packaging.
JAN-B-121	- Barrier-Materials, Greaseproof.

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JAN-P-127 - Packaging and Packing for Overseas Shipment - Tape, Adhesive, Pressure-Sensitive, Water-Resistant.
MIL-B-131 - Barrier-Material; Water-Vaporproof, Flexible.
MIL-D-3464 - Desiccant (Activated) in Bags; for Static Dehumidification and Packaging.
MIL-A-5092 - Adhesive, Rubber (Synthetic and Reclaimed Rubber Base).
MIL-R-6855 - Rubber, Synthetic, Sheet, Molded and Extruded, for Aircraft Application.
MIL-E-10687 - Enamel; Lustreless, Quick-drying (for Ammunition).

STANDARD
Military

MIL-STD-129 - Marking for Shipment and Storage

DRAWING

Army-Navy Aeronautical

AN 7513 - Humidity Indicator, Small.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Level A - Military

3.1.1 Disassembly. - The following components shall be disassembled from the motor drive:

Item	Part No.
Gasket	5589824
Handwheel Assembly	6593727
Shaft	6312635
Spacers	5396835
Pin	544147
Pin	544185

3.1.2 Cleaning. - The exterior unpainted surfaces of the motor drive shall be cleaned in accordance with process C-1 of Specification MIL-P-116, and the other exposed unpainted surfaces of the disassembled components and mounting hardware shall be cleaned by process C-3 of Specification MIL-P-116.

3.1.3 Preservation. - The cleaned unpainted surfaces of the disassembled components and mounting hardware except the gasket, shall be coated with preservative P-5 conforming to Specification MIL-P-116.

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3.1.4 Packaging. - The motor drive shall be prepared for military shipment in accordance with method IIa of Specification MIL-P-116 and as follows: (See figure 2).

3.1.4.1 A wooden base with skids shall be fabricated and the lumber used shall conform to group I or II of Specification JAN-P-106. Carriage bolts, conforming to Specification FF-B-571 shall be placed through the bottom of the skid base. A layer of fiberboard conforming to compliance symbol V3s of Specification JAN-P-108 shall be positioned over the four 3/8 inch diameter bolts. (See figure 1).

3.1.4.2 A gasket fabricated from material conforming to class II, grade 60 of Specification MIL-R-6855 shall be placed on top of the fiberboard pad. The gasket shall be coated on both sides around the bolt holes with rubber adhesive conforming to type 3 of Specification MIL-A-5092.

3.1.4.3 A flexible barrier-material conforming to class 1 of Specification MIL-B-131 shall be positioned over the four mounting bolts in the base and a gasket similar to that specified in 3.1.4.2 with adhesive, shall be placed over the bolts onto the flexible barrier-material.

3.1.4.4 A layer of flexible single face corrugated fiberboard shall be positioned over the bolts. The corrugated fiberboard shall conform to type II or III of Specification LLL-F-291. A single layer of greaseproof barrier material conforming to type I, grade A, class 1 of Specification JAN-B-121 shall be placed over the bolts and on top of the corrugated fiberboard.

3.1.4.5 The motor drive shall be mounted upon the prepared base and secured by the four carriage bolts. A flat washer and lock washer shall be used under each nut. This operation shall be completed before the gasket adhesive dries. The exposed threads above the nuts shall be coated with unthinned paint to prevent the nuts from backing off.

3.1.4.6 The flexible joint, folded to its shortest length, and all other projections of the motor drive shall be covered with greaseproof barrier material conforming to type II, grade A, any class of Specification JAN-B-121. The projections shall be overcovered with cushioning material conforming to type III, class B of Specification UU-C-843. The coverings shall be secured with pressure-sensitive tape conforming to type IV, grade C of Specification JAN-P-127.

3.1.4.7 The preserved spacers and pins shall be wrapped in greaseproof barrier material conforming to type II, grade A, any class of Specification JAN-B-121, and shall be placed into a greaseproof bag conforming to type I, grade A, class C of Specification JAN-P-117. The bag shall be closed by heat sealing after the excess entrapped air is removed.

3.1.4.8 The preserved mounting hardware shall be wrapped and bagged in the same manner as specified in 3.1.4.7.

3.1.4.9 The bagged components and mounting hardware specified in 3.1.4.7 and 3.1.4.8 shall be secured to the terminal assembly housing of the motor drive with pressure-sensitive tape conforming to type II or III, grade C of Specification JAN-P-127.

3.1.4.10 The required units of desiccant conforming to class 2 of Specification MIL-D-3464 shall be secured to the motor drive with twine or pressure-sensitive tape conforming to type II or III, grade C of Specification JAN-P-127. A humidity indicator conforming to Drawing AN7513 shall be secured to the top of the motor drive with pressure-sensitive tape. A protective piece of greaseproof barrier material, conforming to type II, grade A, any class of Specification JAN-B-121 shall be inserted between the desiccant, humidity indicator and motor drive.

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3.1.4.11 The flexible corrugated fiberboard shall be folded around the motor drive. The water-vaporproof barrier shall be closed by heat sealing after the excess entrapped air is removed. The time lapse between securing the desiccant and the closure of the barrier shall be held to an absolute minimum.

3.1.4.12 A piece of greaseproof barrier material conforming to type II, grade C, any class of Specification JAN-B-121 shall be placed over the two 1/2 inch diameter bolts. The cleaned and preserved handwheel assembly shall be mounted on the base and secured by the carriage bolts. A flat washer and lock washer shall be used under each nut. The exposed threads above the nuts shall be coated with unthinned paint or preservative P-1 conforming to Specification MIL-P-116 to prevent the nuts from backing off. The flexible joint of the handwheel assembly shall be folded to its shortest length and covered with greaseproof barrier material conforming to type II, grade C, any class of Specification JAN-B-121. The covering shall be secured with pressure-sensitive tape conforming to type II or III, grade C of Specification JAN-P-127.

3.1.4.13 The gasket (5589824) shall be placed between two pieces of 125 lb. test fiberboard conforming to Specification LLL-B-631. The board shall be secured with gummed tape conforming to class 2 of Specification UU-T-111. The sandwiched gasket shall be inserted in a bag of material conforming to class 2 of Specification MIL-B-131. The bag shall be closed by heat sealing after the excess entrapped air is removed.

3.1.4.14 The preserved surfaces of the shaft shall be covered with greaseproof barrier material conforming to type II, grade C, any class of Specification JAN-B-121. The wrap shall be sealed with pressure-sensitive tape conforming to type II or III, grade C of Specification JAN-P-127.

3.1.5 Packing

3.1.5.1 A cover shall be constructed consisting of ends, sides and top (fabricated according to figure 1). When the cover is assembled to the base it will form a modified style 2 box of the reusable type. The lumber used shall conform to Specification JAN-P-106, group I or II.

3.1.5.2 The bagged gasket shall be secured to the inside of the top panel with pressure-sensitive tape conforming to type II or III, grade C of Specification JAN-P-127.

3.1.5.3 The preserved and wrapped shaft shall be secured to the inside of the top panel by two metal clamps. The shaft shall run diagonally from one through cleat to one filler cleat. The inside surfaces of the metal clamps shall be covered with greaseproof barrier material conforming to type II, grade C, any class of Specification JAN-B-121. The metal clamps shall be secured to the top cover cleats with screws conforming to style 1S of Specification FF-S-111.

3.1.5.4 The cover shall be assembled to the base with lag screws.

3.1.5.5 The exterior surfaces of the reusable container (except the bottom) shall be painted with one coat of quick-drying lusterless enamel conforming to Specification MIL-E-10687.

3.1.5.6 The container shall be strapped in accordance with Appendix 1 of Specification JAN-P-106, except that straps shall not be stapled on domestic shipments. Two straps shall run parallel to the skids with the outer edges of the straps against the inner edges of the skids. One strap shall be applied girthwise around the side and end panels.

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3.1.6 Marking

3.1.6.1 Marking shall be in accordance with Standard MIL-STD-129 and the following special markings.

3.1.6.2 The packaged disassembled components and mounting hardware shall be identified by the quantity and nomenclature of its contents as follows:

1 GASKET
1 HANDWHEEL ASSEMBLY
1 SHAFT
2 SPACERS AND 2 PINS
1 SET OF MOUNTING HARDWARE

3.1.6.3 The serial number of each motor drive shall appear with the permanent marking on the exterior container.

3.1.6.4 The method of preservation (method IIa) shall be included in the basic marking of the shipping container. A method II label shall be attached to the barrier inclosed unit within the shipping container.

3.1.6.5 Each reusable container shall be marked on the exterior, REUSABLE CONTAINER - DO NOT DESTROY. This marking shall also appear on the interior of the top panel.'

3.1.6.6 Oversea marking. - The following marking requirements shall apply to oversea shipments only.

3.1.6.6.1 The Ordnance symbol shall be stenciled on the same surface immediately adjacent to the oversea address. The symbol shall be of a size commensurate with the oversea address and of the same color.

3.1.6.6.2 The SNL group (SNL-F002) shall be stenciled on the sides of the container adjacent to the service color corner marking. The size shall be commensurate with the basic marking on the shipping container.

3.1.6.6.3 Exterior container markings shall be protected with a coating of transparent waterproof material in accordance with Standard MIL-STD-129.

3.2 Level B - Limited Military. - Requirements are the same as specified in 3.1.

3.3 Level C - Minimum Military. -

3.3.1 Disassembly and cleaning. - The disassembly and cleaning of the motor drive shall be the same as specified in 3.1.1 and 3.1.2.

3.3.2 Preserving. - The motor drive shall be preserved to meet the conditions existing during shipment.

3.3.3 Packaging. - All unpainted metal surfaces of the motor drive, disassembled components or mounting hardware shall be covered with greaseproof barrier material.

3.3.4 Packing. - A suitable semi-open skid base shall be prepared and the motor drive shall be mounted on the skid base with carriage bolts. The mounted motor drive shall be enclosed with a suitable domestic wooden box, wirebound crate or three way corner crate in accordance with commercial practice which shall afford complete protection to its contents during transportation by a common carrier at the lowest reasonable cost.

3.3.5 Marking. - Marking shall be in accordance with Standard MIL-STD-129 and the exterior container shall be marked with letters 3/4 inch high as follows: "MIN MIL PK."

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3.4 Workmanship. - All operations and processes involved in accomplishing the cleaning, preserving, packaging, packing and marking requirements specified herein shall be in accordance with the highest grade practices. The finished pack shall be free from defects that may affect its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Pilot package. - A preproduction pilot package, type of shipment as specified by the procuring activity, shall be in accordance with this Specification. The pilot package shall be inspected and approved by a Government inspector.

4.2 Inspection. - Inspection of the entire packaging process shall be made. For each shipment of motor drives, the shipping container and the contents shall be thoroughly examined to assure compliance with all the details of this specification.

4.3 Test procedures. - At the discretion of the Government inspector, any of the applicable tests of Specification JAN-P-100 and MIL-P-116 shall be required to assure compliance with this specification.

5. PREPARATION FOR DELIVERY

5.1 There are no applicable requirements

6. NOTES

6.1 Ordering data. - Procurement documents should specify the title, number, and date of this Specification, and the type of shipment to be made.

6.2 The following indicates the stock number to be added in the identification data:

Item	Stock No.
Drive, Motor, M2A2	F002-7652333

6.3 The following tables are furnished for information only:

TABLE I
*DIMENSIONS OF SHIPPING CONTAINER

Military Shipment	Dimensions in Inches		
	Length	Width	Height
Inside	27 3/4	22	13
Outside	29	23 1/2	16 5/8
Displacement		6.6 cubic feet.	

TABLE II
*WEIGHTS

	Military Shipment
Net	150 lbs.
Tare	85 lbs.
Gross	235 lbs.

*Container dimensions and weights are approximate and may vary slightly due to manufacturer's practice.

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TABLE III
BILL OF MATERIALS
Dimensions in Inches (Min.)
Military Shipment

No.	Part	Req'd	Length	Width	Thickness	Reference
1	Skid	1	29	3 5/8	1 5/8	JAN-P-106 gp 1 or II
2	Skid	1	29	7 5/8	1 5/8	JAN-P-106 gp I or II
3	Floor Boards	1	27 3/4	22	5/8	JAN-P-106 gp I or II
4	Carriage Bolts	4	4 1/2	3/8 dia with flat washer, lock washer and nut per bolt		FF-B-571
5	Carriage Bolts	2	4 1/2	1/2 dia. with flat washer, lock washer, and nut per bolt		FF-B-571
	Nails, Flooring 8d	As required				FF-N-105
6	*Side Panels	2				JAN-P-106 gp I or II
	Sheathing	2	23 1/2	14 3/8	5/8	
	Through Cleats	4	13	3 5/8	3/4	
	Filler Cleats	4	13 1/4	3 5/8	3/4	
7	*End Panels	2				JAN-P-106 gp I or II
	Sheathing	2	27 3/4	15 3/4	3/4	
	Through Cleats	4	27 3/4	3 5/8	3/4	
	Filler Cleats	4	5 3/4	3 5/8	3/4	
8	*Top Panel	1				JAN-P-106 gp I or II
	Sheathing	1	29	23 1/2	5/8	

*Container dimensions are approximate and may vary slightly due to manufacturer's practice.

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No.	Part	Req'd	Dimensions in Inches (Min.) Length	Width	Thickness	Reference
	Through Cleats	2	27 3/4	3 5/8	3/4	
	Filler Cleats	2	14 3/4	3 5/8	3/4	
	Nails, 6d	As required for cleats				FF-N-105
	Nails, 8d	As required for assembly of box				FF-N-105
	Cleaning Material	As required				MIL-P-116 C-1 and C-3
	Preservative	As required				MIL-P-116 P-5
9	Fiberboard	1	17	9		JAN-P-108 V3s
10	Gaskets	2	17	9		MIL-R-6855 cl II, gr. 60
11	Adhesive	As required				MIL-A-5092 type 3
12	Barrier Material		55	50		MIL-B-131 cl 1
13	Fiberboard, Single Face		50	45		LLL-F-291 type II or III
14	Greaseproof Barrier Material		17	9		JAN-B-121 type I, gr A, cl 1
15	Desiccant		2.5 lbs (40 units)			MIL-D-3464 cl 2
16	Humidity Indicator	1				AN 7513
	Greaseproof Barrier Material	As required				JAN-B-121 ty II gr A any class
	Cushioning Material	As required				UU-C-843 type III, cl B
	Tape, Pressure-Sensitive	As required				JAN-P-127 type II or III, gr C
	Unthinned Paint or Preservative	As required As required				MIL-P-116 P-1
17	Greaseproof Barrier Material		16	6		JAN-B-121 type II, gr C any class

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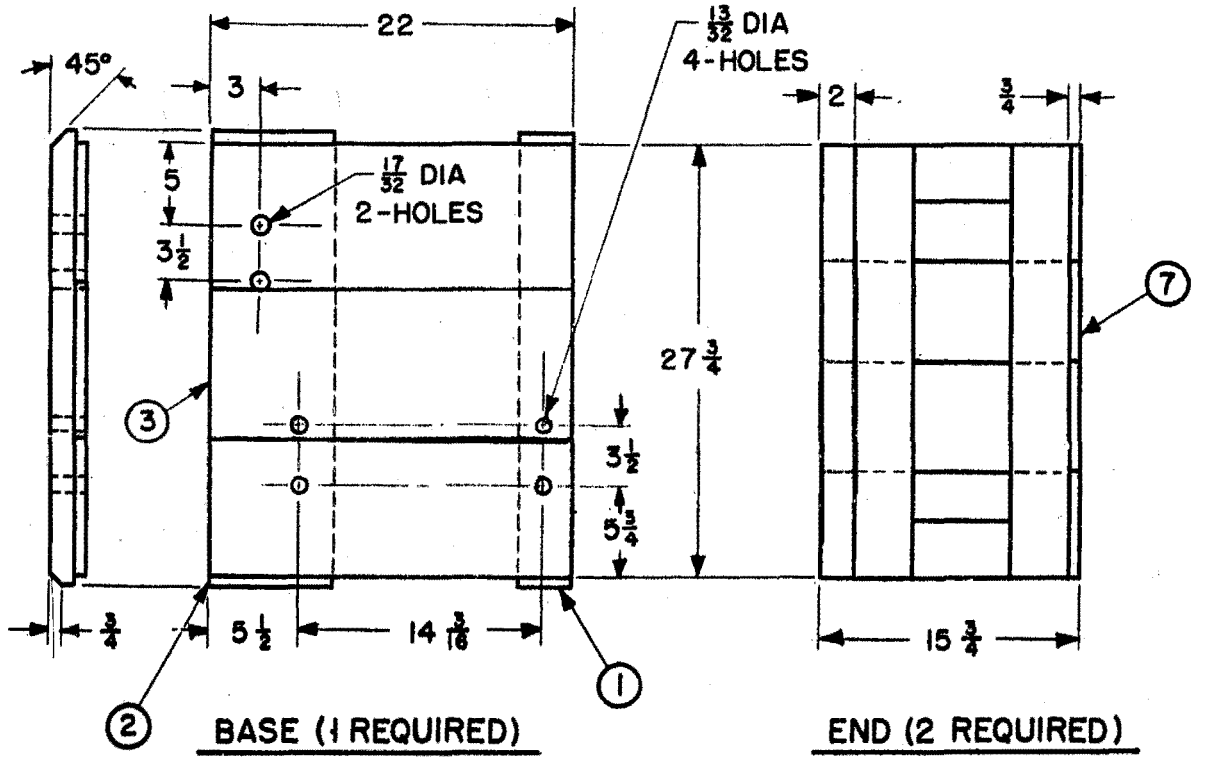
No.	Part	Req'd	Dimensions in Inches (Min.) Length	Width	Thickness	Reference
18	Bag, Greaseproof	1	4	3		JAN-P-117 type I, gr A, cl C
19	Bag, Greaseproof	1	6	4		JAN-P-117 type I, gr A, cl C
20	Clamps, Metal	2	Standard 3/4 inch radius pipe clamp with single hole mounting.			
	Screws	2	As required for securing clamps			FF-S-111 style 1s
	Fiberboard	2	15 1/4	8 1/2		LLL-B-631,125 lb.
	Tape		8	1		UU-T-111, cl 2
21	Bag, Moisture-proof	1	16	9		MIL-B-131 cl II
22	Strapping, Flat		290	3/4	.020	QQ-S-781 type 1, cl A or B
	or Round		290	.1055 (12 ga.)		QQ-S-790 cl. A or B.
	Staples		As required			FF-N-105

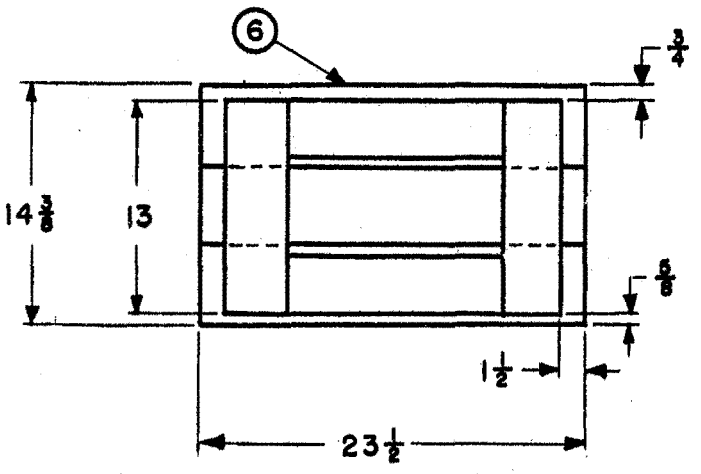
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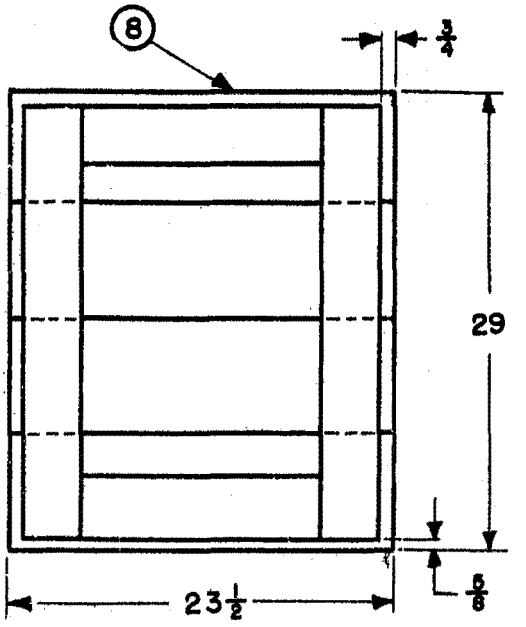


FIGURE 1

DRIVE, MOTOR, M2A2
BOX FOR MILITARY SHIPMENT

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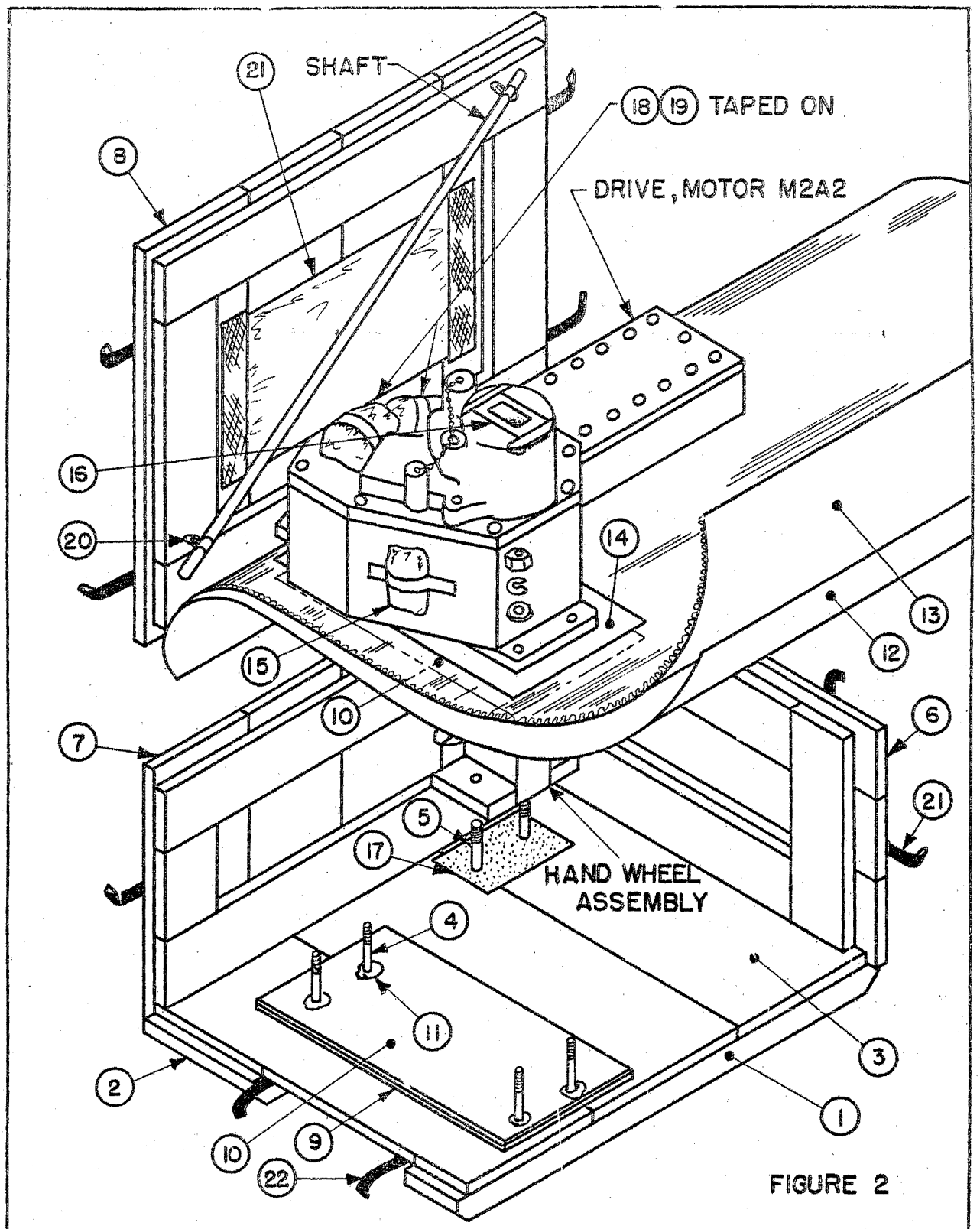


FIGURE 2

DRIVE, MOTOR, M2A2, PACKED FOR MILITARY SHIPMENT

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